MIS 600 · DATABASE MANAGEMENT SYSTEMS

Chapter 2

Modeling Data in the Organization

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Fall 2026

Text: Modern Database Management, 14th ed. (Hoffer, Topi, Venkataraman, and Bala, 2026). Figures in this deck come from Chapter 2 and are used for instruction in this course.

The plan for tonight

- First half. Why we model before we build. Business rules. Names and definitions.
- The E-R model. Entities, attributes, and identifiers.
- Break.
- Second half. Relationships, associative entities, and cardinality.
- Time. What changes when history matters.
- Pine Valley Furniture. Reading a complete model, then testing it with live SQL.

Bring up DB Fiddle when we reach the worksheet.

What you should be able to do after this chapter

- Explain why data modeling carries the highest payoff in systems work
- Write good names and definitions for entities, attributes, and relationships
- Read an E-R diagram out loud, in both directions, as business rules
- Model composite, multivalued, and derived attributes, and choose sound identifiers
- Tell apart strong, weak, and associative entities, and unary, binary, and ternary relationships
- State minimum and maximum cardinalities, and convert a many-to-many relationship into an associative entity
- Model simple time-dependent data with time stamps

PART ONE

Rules of the business

Why we model before we build

- The model drives everything. Tables, programs, and integrity checks all come from the facts and rules captured here.
- Data are the complex part. Most modern systems live or die by their data, not their process flow.
- Data outlive processes. How WSU registers students changes every few years. Data about students and courses go back a century.
- Models are cheap to change. A wall moved on paper costs an eraser. Moved after the concrete is poured, it costs a fortune.

“Measure twice, cut once.”

Seven deadly sins of database design

Embarcadero Technologies (2014), as cited in the text. Chapter 2 goes straight at three of them.

The sin	What it looks like	Where we fix it
Poor or missing documentation	Nobody wrote down what the data mean, so every developer guesses	Ch. 2 and 3
Little or no normalization	Facts repeated in the wrong places, updates go wrong	Ch. 4
Model treated as finished	Drawn once, never revisited as requirements change	All semester
Reference data stored badly	Lookup values scattered or hard-coded	Parts II and III
No foreign keys or check constraints	The database cannot defend its own rules	Ch. 4 and 8
No domains or naming standards	Every developer names things differently	Ch. 2
No primary keys	Rows cannot be told apart reliably	Ch. 2, 4, and 8

If the data are wrong, nothing built on top of them can be right.

Business rules

A business rule is a statement that defines or constrains some aspect of the business.
GUIDE Business Rules Project (1997), as cited in the text

- "A student may register for a section only after completing its prerequisites."
- "A preferred customer gets a 10 percent discount unless the account is overdue."

- Rules govern how data get created, updated, and removed
- In our scope only if it touches data. "Friday is business casual" is real policy, but not our problem.
- Businesspeople own and approve the rules. The analyst surfaces them and sharpens the vague ones.

Data modeling is documenting the rules of the business that govern data.

What makes a rule good

- Declarative. States policy, not the procedure that enforces it
- Precise. One interpretation only. “Large orders need approval” fails: how large?
- Atomic. One statement, complete on its own
- Consistent. No contradictions with other rules
- Expressible. Can be written in structured plain language
- Distinct. Not a rephrased duplicate
- Business-oriented. In words businesspeople understand

Characteristic	Explanation
Declarative	A business rule is a statement of policy, not how policy is enforced or conducted; the rule does not describe a process or implementation but rather describes what a process validates.
Precise	With the related organization, the rule must have only one interpretation among all interested people, and its meaning must be clear.
Atomic	A business rule marks one statement, not several; no part of the rule can stand on its own as a rule (i.e., the rule is indivisible, yet sufficient).
Consistent	A business rule must be internally consistent (i.e., not containing conflicting statements) and must be consistent with (and not contradict) other rules.
Expressible	A business rule must be able to be stated in natural language, but it will be stated in a structured natural language so that there is no misinterpretation.
Distinct	Business rules are not redundant, but a business rule may refer to other rules (especially to definitions).
Business-oriented	A business rule is stated in terms businesspeople can understand, and because it is a statement of business policy, only businesspeople can modify or invalidate a rule; thus, a business rule is owned by the business.

Keep asking: Always true? Special cases? One or many? Keep history?

Naming data objects

Salin (1990), as cited in the text. Write the definition first, then build the name from it.

A good name is	Good	Poor
About the business, not the technology	Customer	File10, Bit7
Meaningful, nearly self-documenting	Order Date	It, Has, Person
Unique among all data objects	Home Address / Campus Address	Address
Readable, the way people say it	Grade Point Average	Average Grade Relative To A
Built from approved words	Maximum	Ceiling, Upper Limit
Repeatable by different people	Student Birth Date	Emp DOB
In a standard order	Employee Hire Date	Date Hired Emp

Defining data objects

- A term is a word with a specific business meaning: course, flight, reservation
- A fact links terms in one declarative sentence: “A course is a module of instruction in a particular subject area.”
- Say what the thing is, when instances are created and deleted, and what history is kept
- Constraints are separate rules, not clauses buried inside a definition
- Harder than it sounds. Who counts as a “student”? Prospect, applicant, matriculated, former, graduate. Chapter 3 gives us subtypes for exactly this problem.

“The person who controls the meaning of data controls the data.”

PART TWO

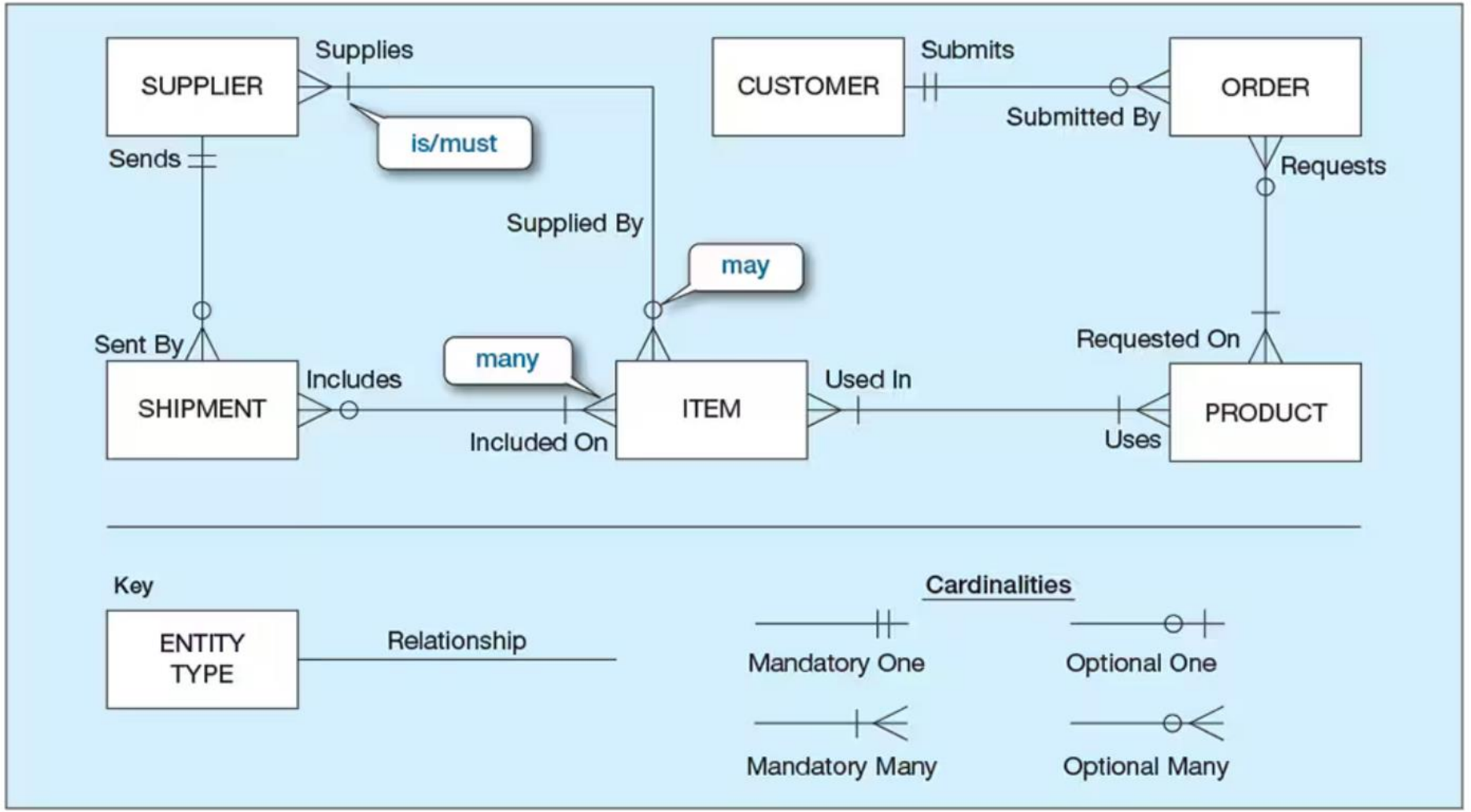
The E-R model: entities and attributes

The entity-relationship model

- E-R model. A logical picture of the data of a business: entities, relationships among them, and attributes of both (Chen, 1976)
- E-R diagram (ERD). The drawing of that model
- Conceptual. It describes the business, not any DBMS. Normalization and foreign keys wait for Chapter 4.
- Our main tool for talking with users during analysis
- No single standard notation exists. The text blends what professional tools use, with Microsoft Visio shown for comparison. Draw.io reads like Visio.

Our first full diagram: Pine Valley Furniture

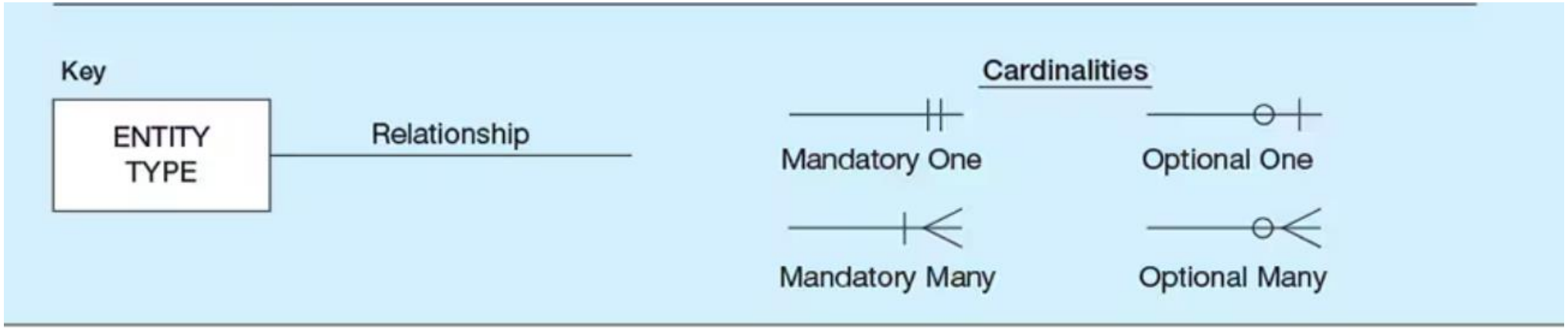
Figure 2-1. Suppliers ship items, items build products, customers order products. Attributes omitted.



How to read the marks on the lines

The single most common mistake in week one lives on this slide.

- Walk to the far end. The marks next to ORDER answer: how many orders for one customer?
- Circle = zero allowed (optional). Bar = at least one (mandatory). Crow's foot = many. Single bar = one.
- The grammar. <entity> <min> <relationship> <max> <entity>, so: <CUSTOMER> <may> <Submit> <any number> <ORDER>
- Read every diagram aloud, both directions, before you trust it



The definitions under the diagram are metadata

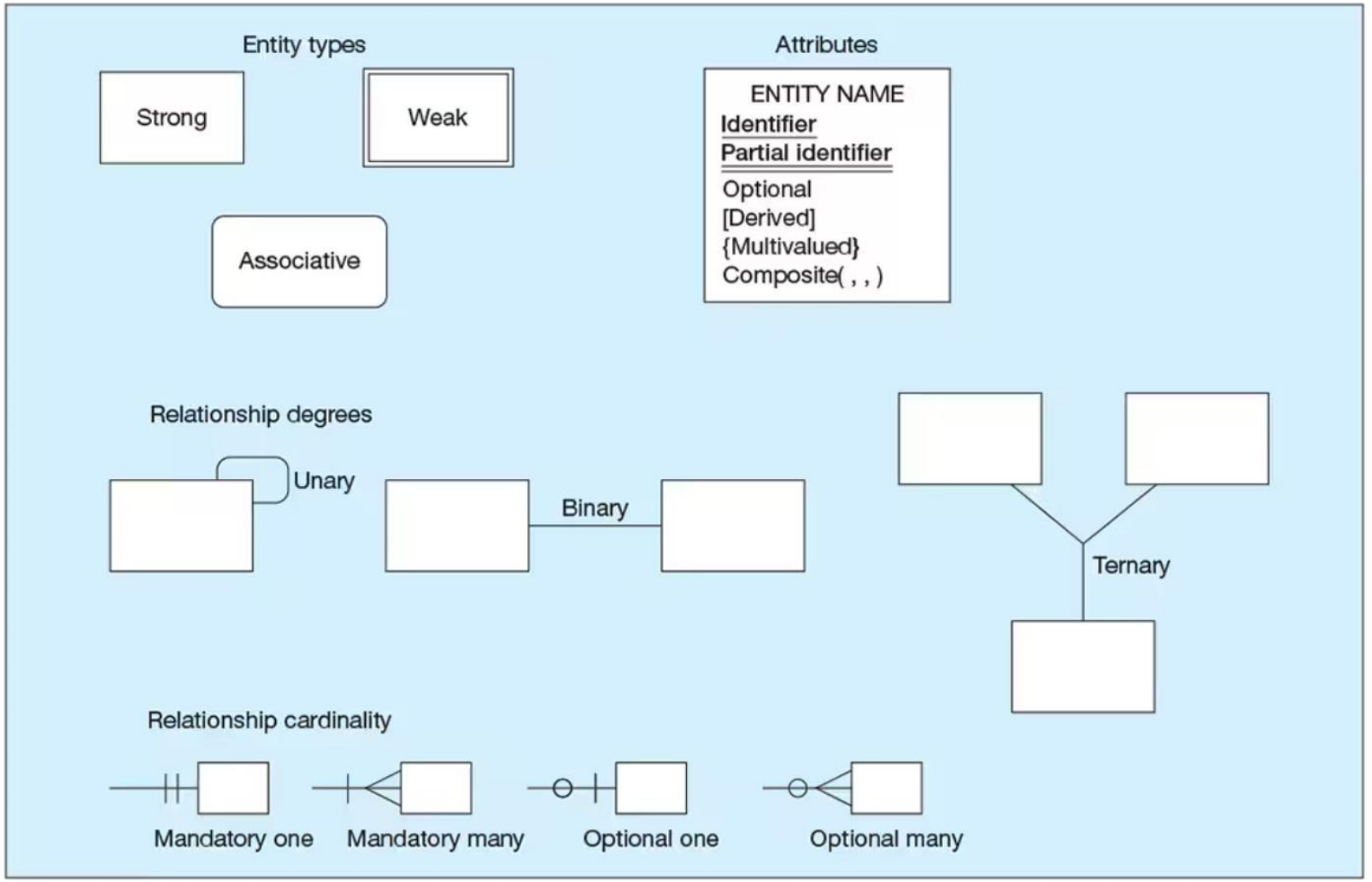
The written half of Figure 2-1. The picture alone is not the model.

- Every entity gets a written definition, agreed on by the business
- Homonyms. Accounting's "customer" has paid an invoice. Marketing's has merely been contacted. Same word, different data.
- Here a CUSTOMER includes people who might order, so some customers have no orders. That is why Submits is optional.
- A clean diagram over fuzzy definitions still misleads

CUSTOMER	A person or an organization that has ordered or might order products. *Example*: L. L. Fish Furniture.
PRODUCT	A type of furniture made by Pine Valley Furniture that may be ordered by customers. Note that a product is not a specific bookcase because individual bookcases do not need to be tracked. *Example*: A 6-foot, 5-shelf, oak bookcase called O600.
ORDER	The transaction associated with the sale of one or more products to a customer and identified by a transaction number from sales or accounting. *Example*: The event of L. L. Fish buying one product O600 and four products O623 on September 10, 2024.
ITEM	A type of component that goes into making one or more products and can be supplied by one or more suppliers. *Example*: A 4-inch ball-bearing caster called I-27–4375.
SUPPLIER	A company that may provide items to Pine Valley Furniture. *Example*: Sure Fasteners, Inc.
SHIPMENT	The transaction associated with items received in the same package by Pine Valley Furniture from a supplier. All items in a shipment appear on one bill-of-lading document. *Example*: The receipt of 300 I-27-4375 and 200 I-27-4380 items from Sure Fasteners, Inc., on September 9, 2024.

The notation we use all semester

Figure 2-2. Keep this page open while you draw. Chapter 3 adds a few more symbols.



Entities: the nouns of the business

- A person, place, object, event, or concept the organization keeps data about
- Events count. A SALE or a REGISTRATION is an entity because we must remember it happened.
- Concepts count. You cannot touch an ACCOUNT, but the business runs on it.
- Singular noun, written in CAPITALS

Person:	EMPLOYEE, STUDENT, PATIENT
Place:	STORE, WAREHOUSE, STATE
Object:	MACHINE, BUILDING, AUTOMOBILE
Event:	SALE, REGISTRATION, RENEWAL
Concept:	ACCOUNT, COURSE, WORK CENTER

Entity type versus entity instance

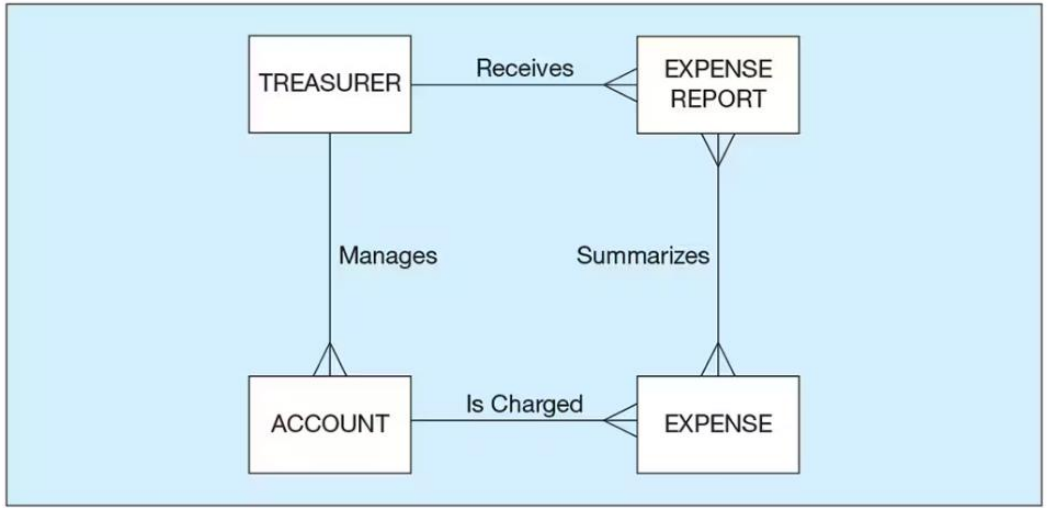
Figure 2-3. The type is the blank form, described once as metadata. Instances are the filled-in copies, stored by the thousand.

Entity type: EMPLOYEE			
Attributes	**Attribute Data Type**	**Example Instance**	**Example Instance**
Employee Number	CHAR (10)	64217836	53410197
Name	CHAR (25)	Michelle Brady	David Johnson
Address	CHAR (30)	100 Pacific Avenue	450 Redwood Drive
City	CHAR (20)	San Francisco	Redwood City
State	CHAR (2)	CA	CA
Zip Code	CHAR (9)	98173	97142
Date Hired	DATE	03-21-2021	08-16-2023
Birth Date	DATE	08-16-1998	09-04-2003

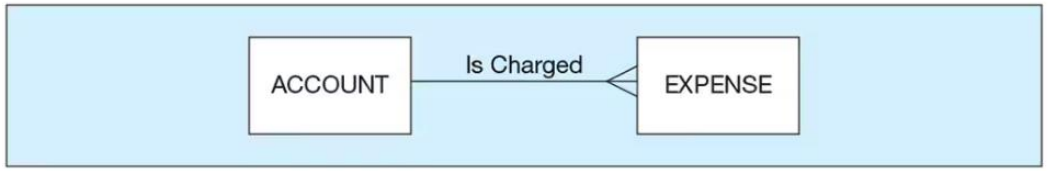
Not everything that moves is an entity

Figure 2-4. A sorority expense system, before and after the cleanup.

- The Treasurer enters the data and reads the reports. She is a user, and there is only one of her. Nothing to store.
- The Expense Report is an output, computed from ACCOUNT and EXPENSE data already in the database
- Verbs like Receives and Summarizes describe data moving, not data related. They are the tell.
- The test. A real entity has many instances, each with a distinguishing characteristic plus other descriptive data.



(a) System user (Treasurer) and output (Expense Report) shown as entities

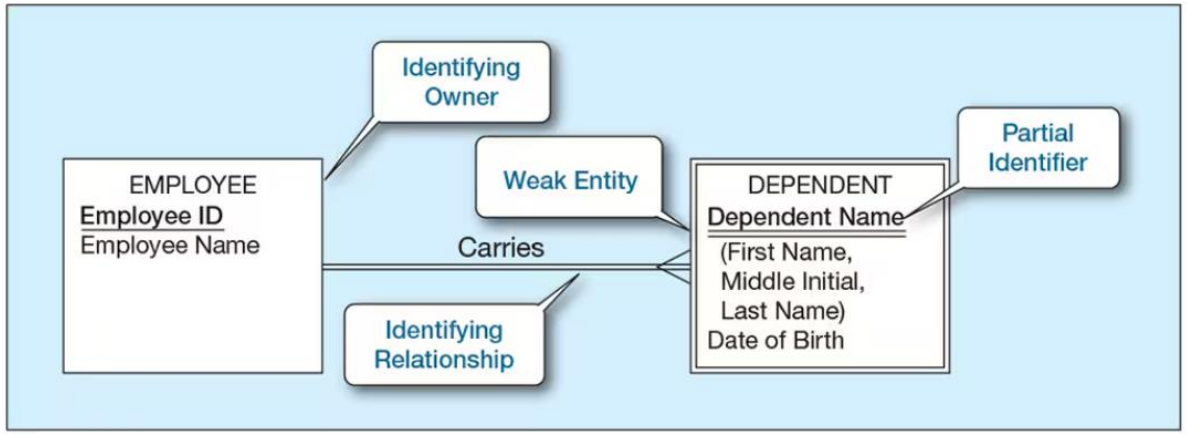


(b) E-R diagram with only the necessary entities

Strong and weak entities

Figure 2-5. Double box, double line, double underline: the whole weak-entity kit.

- Strong. Stands on its own and has its own identifier: EMPLOYEE, COURSE
- Weak. Exists only through its identifying owner, reached by the identifying relationship (Carries)
- Partial identifier. Dependent Name repeats across employees. Add Employee ID and it becomes unique.
- The delete test. Remove the employee and the dependent record has no reason to exist.
- More pairs: BOOK and BOOK COPY, COURSE and COURSE OFFERING, MOVIE and DVD



Attributes: required or optional

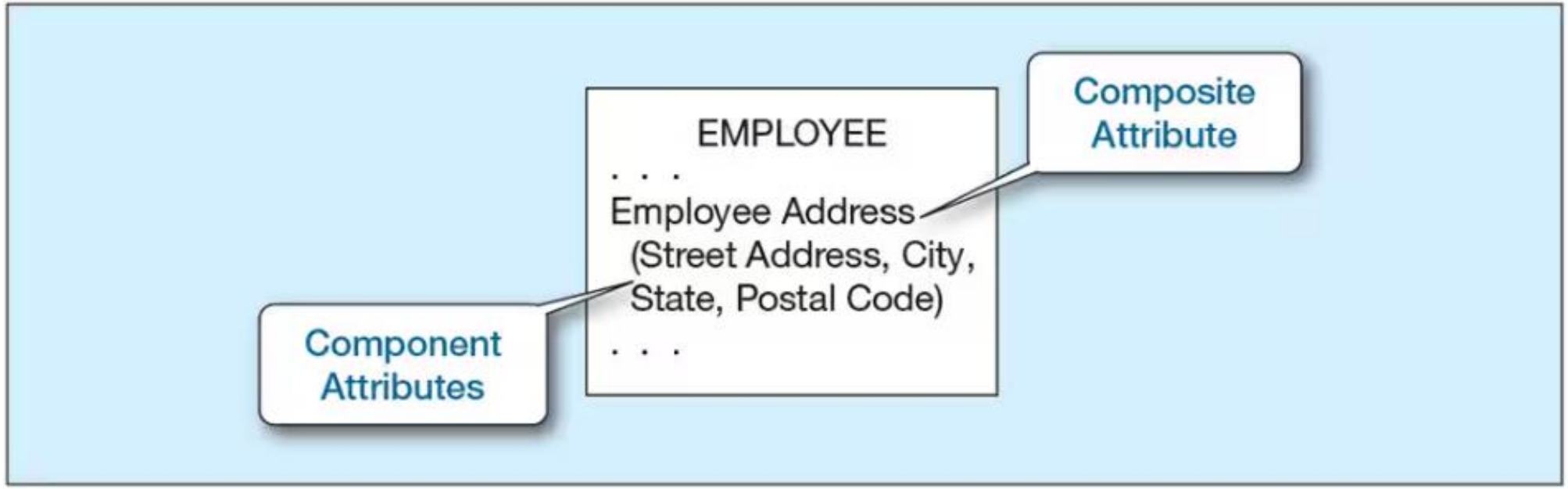
Figure 2-6. Bold means required. Plain means optional. An empty value is null.

- Every instance must supply the bold attributes before it can be stored at all
- Melissa Kraft has no Major yet. The model says that is fine.
- Naming. Initial capitals, spaces between words: Student Home Address
- An entity holding nothing but an identifier is a code list, not an entity
- Attributes stay home: DEPENDENT carries no Employee ID. The relationship already knows.

Entity type: STUDENT				
Attributes	**Attribute Data Type**	**Required or Optional**	**Example Instance**	**Example Instance**
Student ID	CHAR (10)	Required	28-618411	26-844576
Student Name	CHAR (40)	Required	Michael Grant	Melissa Kraft
Home Address	CHAR (30)	Required	314 Baker St.	1422 Heft Ave
Home City	CHAR (20)	Required	Centerville	Miami
Home State	CHAR (2)	Required	OH	FL
Home Zip Code	CHAR (9)	Required	45459	33321
Major	CHAR (3)	Optional	MIS	

Composite attributes

Figure 2-7. One value with meaningful parts.

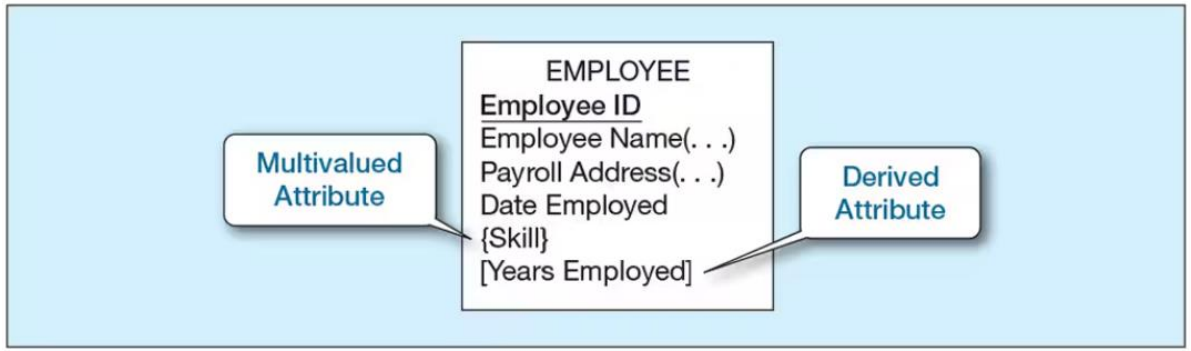


- Users can ask for Employee Address whole, or for City alone
- Split only if someone will ever use the parts separately. Marketing sorts by state, so State earns its own part.
- Most drawing tools have no symbol for this. List the parts in parentheses, as the book does.

Multivalued and derived attributes

Figure 2-8. Curly brackets hold many values. Square brackets hold a calculation.

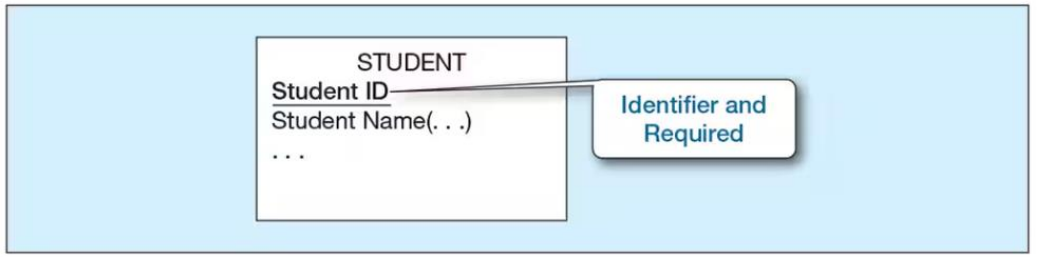
- {Skill}. One employee, several values
- [Years Employed]. Computed from Date Employed and today's date. Store the formula, not the number.
- Do not confuse them. Composite is one value with parts. Multivalued is several values.
- They combine: {Customer Address (Street, City, State, Zip)} for home, office, and seasonal addresses



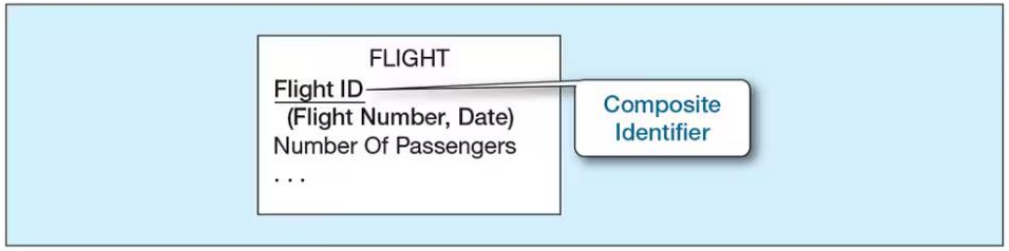
Identifiers

Figure 2-9. Underline the identifier. It is bold too, because it can never be empty.

- The attribute whose value singles out one instance: Student ID, Vehicle ID
- Student Name fails. Names repeat, and people change them.
- Composite identifier. Flight 1204 flies every day. Flight Number plus Date breaks the tie.
- Keep asking the tie-breaker question: could two flights share a number and a date?



(a) Simple identifier attribute



(b) Composite identifier attribute

Choosing among candidate identifiers

Bruce (1992), as cited in the text.

- Stable. The value never changes over the life of the instance
- Always present. Guaranteed valid and never null, every part of it
- Not intelligent. Meaning baked into the digits goes stale when the warehouse moves
- Compact. Prefer a single surrogate (Game Number) to a long composite (Home Team plus Visiting Team)

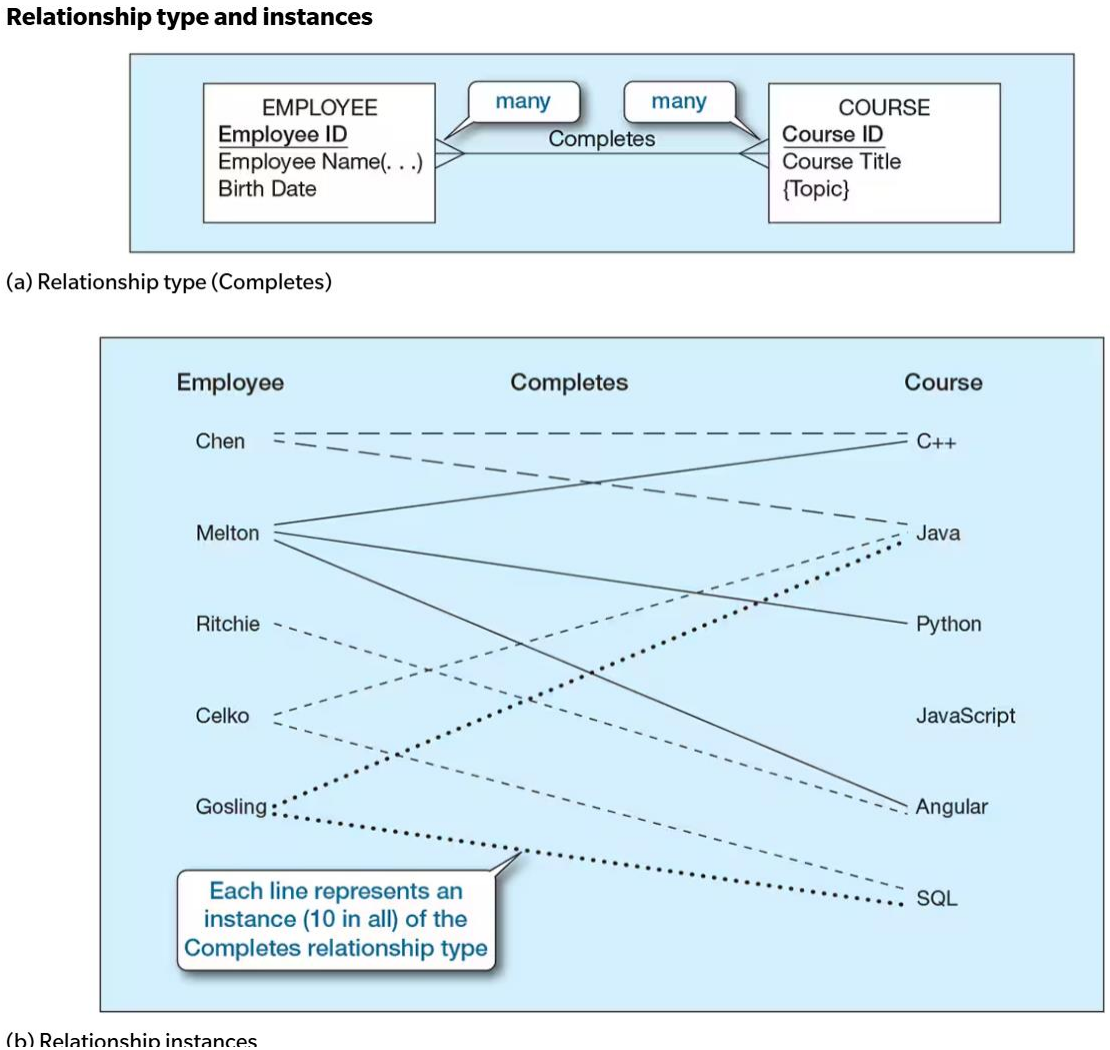
Your myWSU ID passes all four. Your name fails two, and Social Security numbers fail for a different reason: security.

PART THREE

Relationships and cardinality

Relationships: the verbs of the business

Figure 2-10. Completes is the relationship type. Each of the ten lines below is one instance: one employee paired with one course.



Where does Date Completed belong?

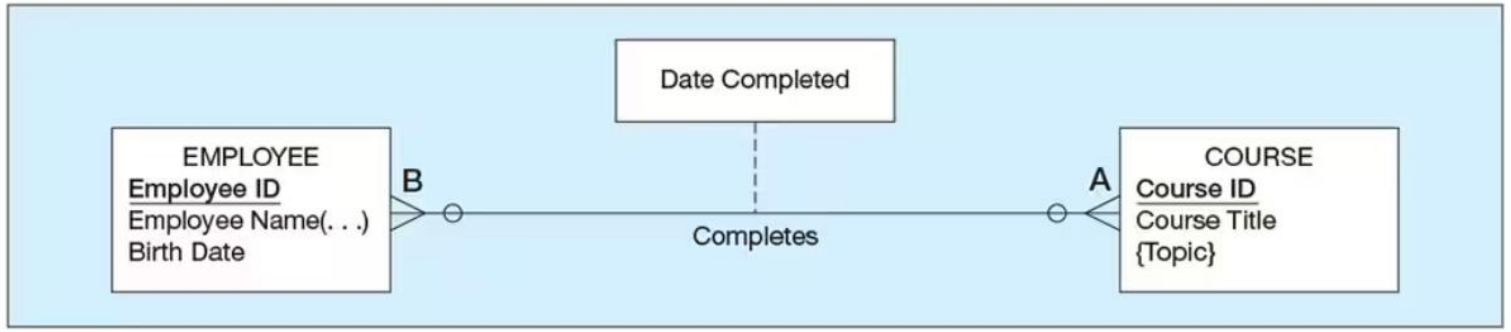
Table 2-2. Not on EMPLOYEE, not on COURSE. It describes the pair.

- Melton has three completion dates, so no single date describes him
- C++ was completed on different dates by different people, so no single date describes the course
- It belongs to the relationship. One value per instance of Completes.
- Allowed on many-to-many and one-to-one only. On one-to-many, the attribute already belongs to the many side: step away from the relationship.
- Book note. Table 2-2 swaps two rows against Figure 2-10 (Melton's Angular, Gosling's SQL). Follow the figure.

EMPLOYEE NAME	COURSE TITLE	DATE COMPLETED
Chen	C++	06/2024
Chen	Java	09/2024
Melton	C++	06/2024
Melton	Python	02/2025
Melton	SQL	03/2024
Ritchie	Angular	11/2024
Celko	Java	03/2024
Celko	SQL	03/2025
Gosling	Java	09/2024
Gosling	Angular	06/2024

A relationship with attributes wants to be an entity

Figure 2-11. Completes becomes CERTIFICATE, with its own identifier and Date Completed. The crow's feet move onto it, and both parents become mandatory one.



(a) Attribute on a relationship



(b) An associative entity (CERTIFICATE)



(c) An associative entity using Microsoft VISIO

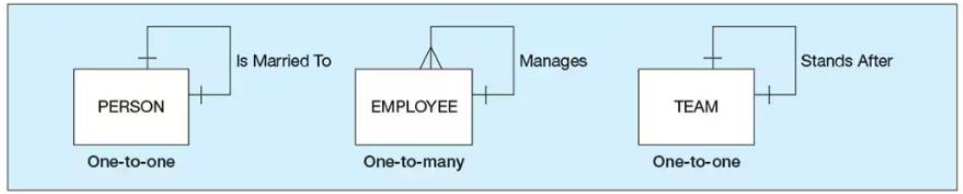
When to convert: four conditions

- Every surrounding relationship is a “many”
- It means something to users on its own, ideally with a one-attribute identifier
- It carries at least one attribute beyond that identifier
- It can enter relationships of its own

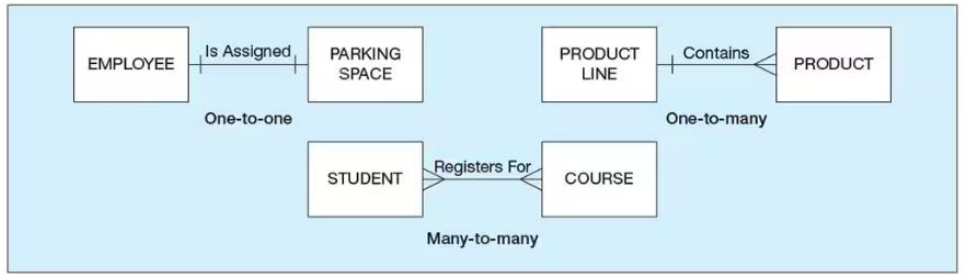
- A marriage. Links two PERSON instances, has its own date, place, and license number, and can relate to other things later. The marriage is the associative entity.
- You have drawn one already. ENROLLMENT between STUDENT and SECTION, carrying Grade, in the Chapter 1 Draw.io starter. ORDER LINE plays the same part at Pine Valley.

Degree: how many entity types take part

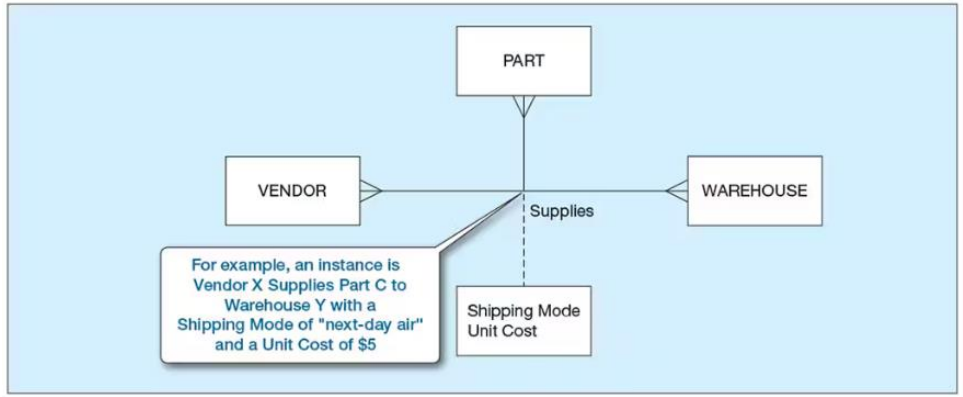
Figure 2-12. Unary loops back on one entity. Binary joins two. Ternary joins three at once. Cardinalities state one organization's rules, not the world's.



(a) Unary relationships



(b) Binary relationships



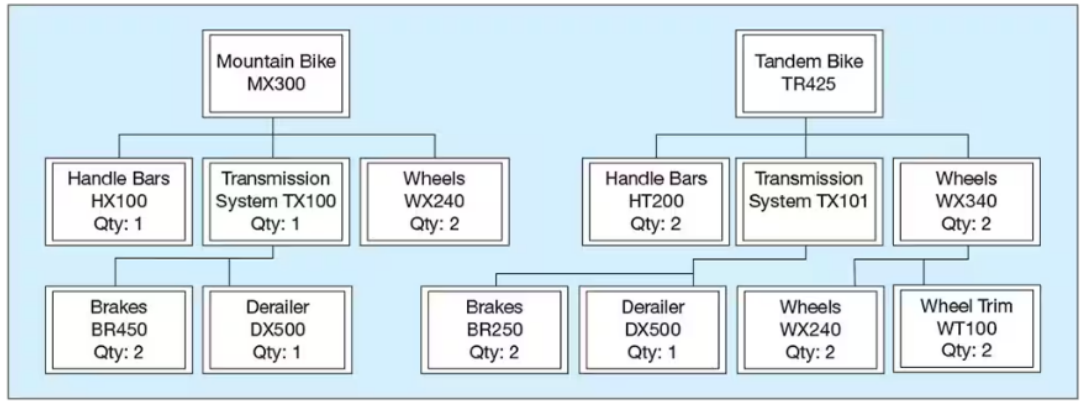
(c) Ternary relationship

The famous unary: a bill of materials

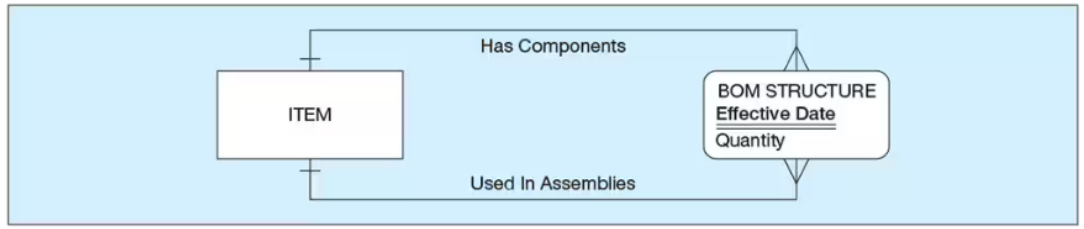
Figure 2-13. Many-to-many on ITEM: WX240 goes into two assemblies, and MX300 has three components. Quantity rides the relationship, so it becomes BOM STRUCTURE.



(a) Many-to-many relationship



(b) Two ITEM bill-of-materials structure instances

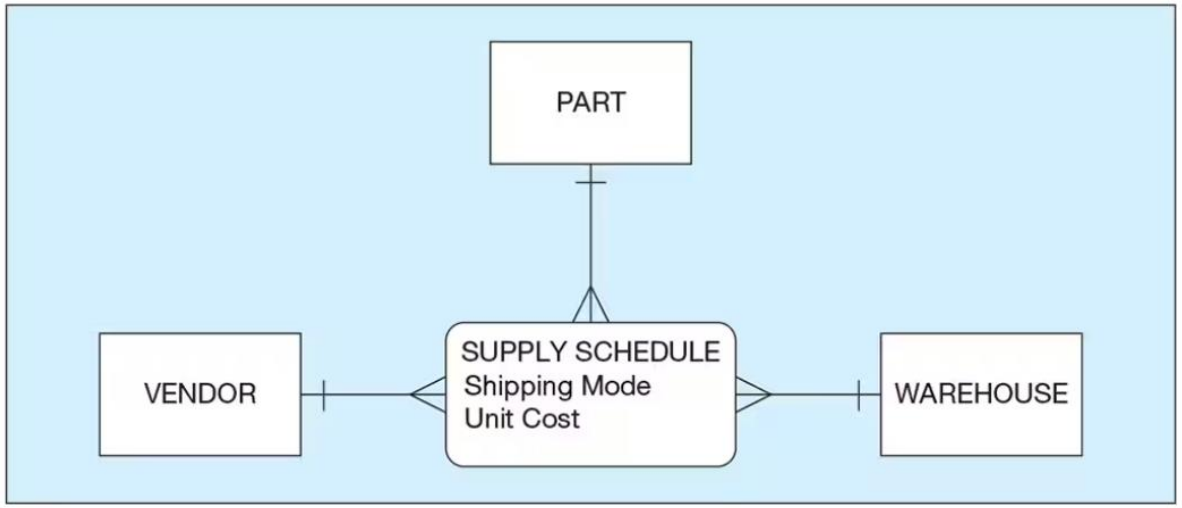


(c) Associative entity

Ternary is not three binaries

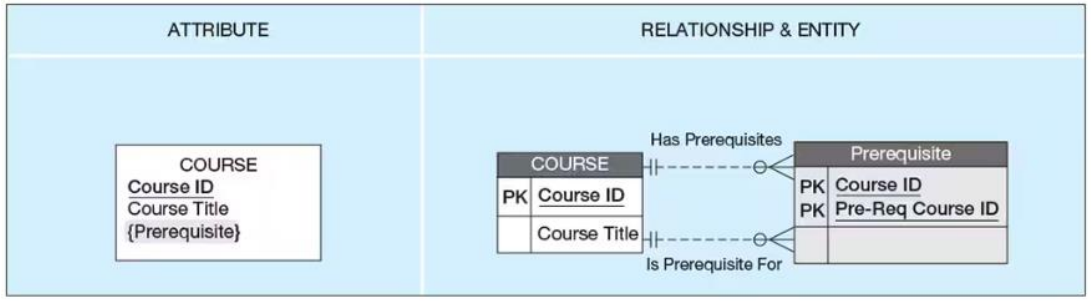
Figure 2-14. The Supplies relationship, rebuilt as an associative entity.

- Unit Cost belongs to the triple: vendor, part, and warehouse together
- “Vendor X ships part C for $8” is an incomplete fact. To which warehouse?
- Split it into three pairs and the fact is gone
- Rule. Convert every ternary to an associative entity. Leave its lines unnamed. The identifier can wait for Chapter 4.
- Most drawing tools cannot draw a true ternary anyway

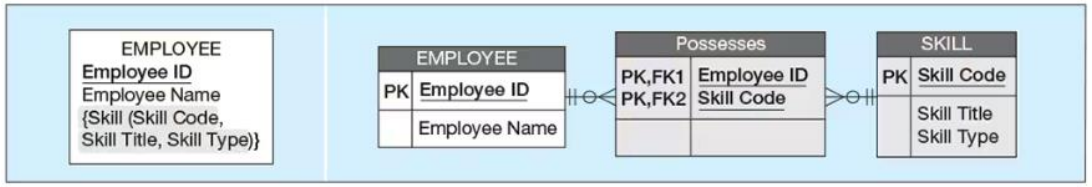


Attribute, or entity of its own?

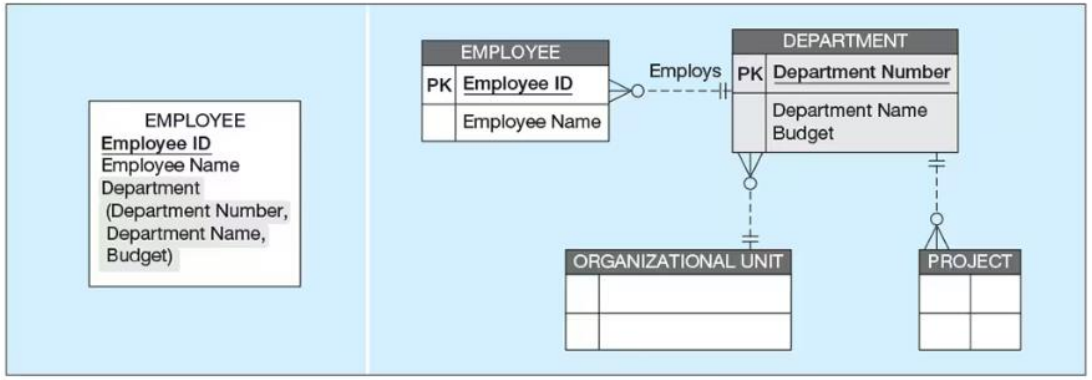
Figure 2-15. Promote it to an entity when the values are shared: many employees hold one SKILL, many things need one DEPARTMENT. The right column is how the tables will look.



(a) Multivalued attribute versus relationships via bill-of-materials structure



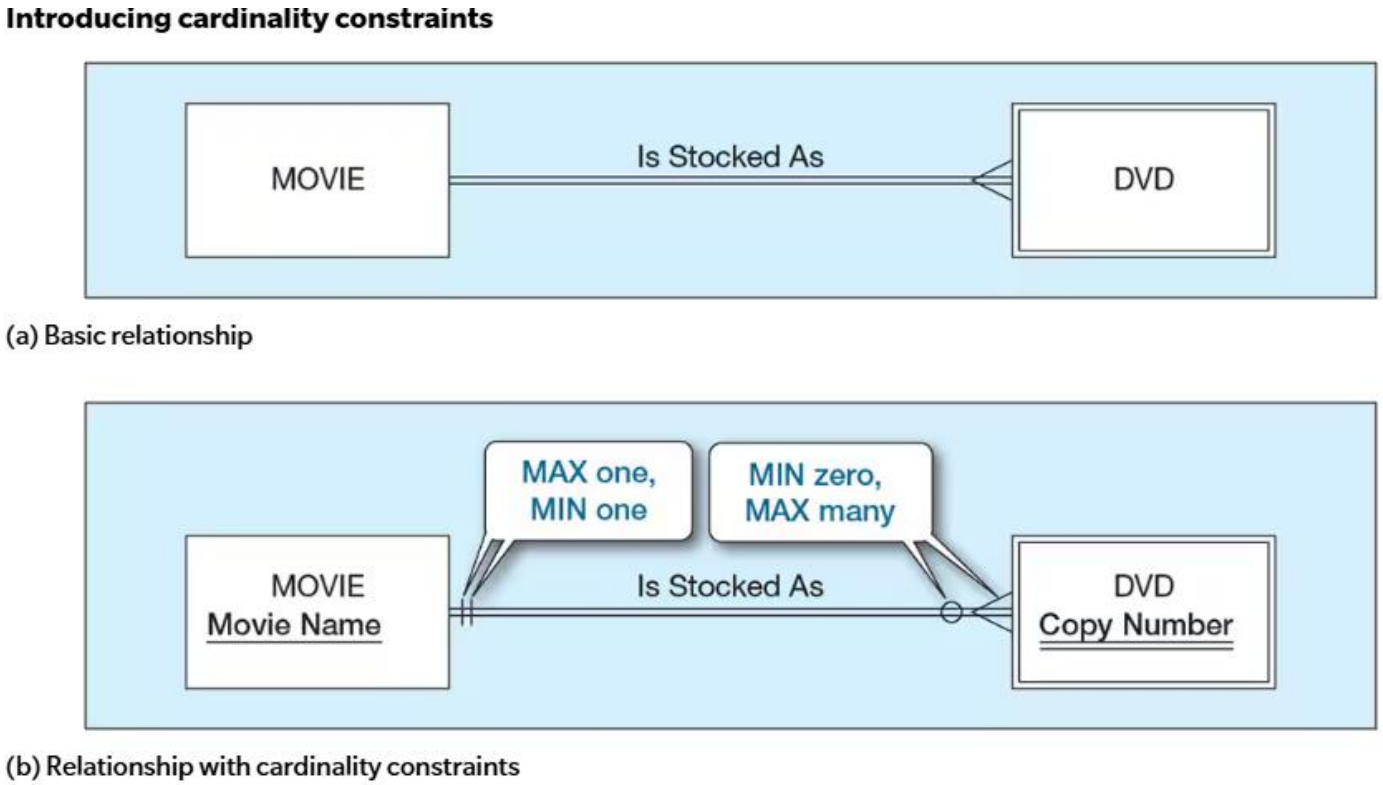
(b) Composite, multivalued attribute versus relationship



(c) Composite attribute of data shared with other entity types

Cardinality constraints

Figure 2-16. The kiosk may hold zero DVDs of a movie tonight. The plain line in (a) cannot say that.



- Minimum. Fewest allowed. Zero means optional, one means mandatory.
- Maximum. Most allowed. One, many, or a fixed number.
- Every DVD copies exactly one MOVIE: mandatory one
- DVD is weak. Copy Number plus Movie Name names one disc.

The four combinations

Reading	Marks at the far end	Minimum	Maximum
Exactly one	two bars	1	1
One or more	bar, then crow's foot	1	many
Zero or one	circle, then bar	0	1
Zero, one, or more	circle, then crow's foot	0	many

Max sits against the box, min sits farther out. *“Max inside, min outside.”*

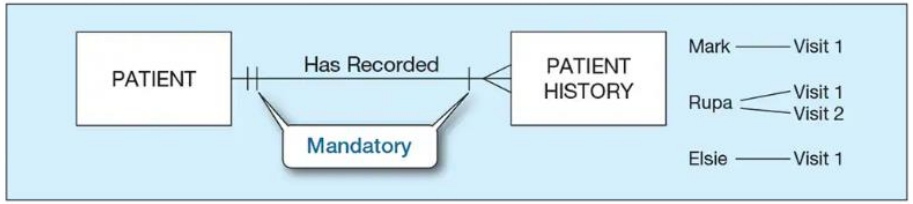
A fixed number can replace many: write 5 at the crow's foot and the rule becomes “at most five.”

Every combination, with live instances

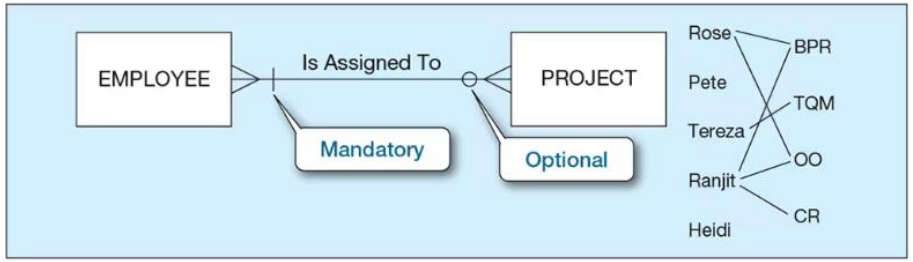
Figure 2-17. Mandatory both ways, mixed, optional both ways. Pete has no project, and that is legal. Cover the left half and reconstruct it from the sentences.

Figure 2-17

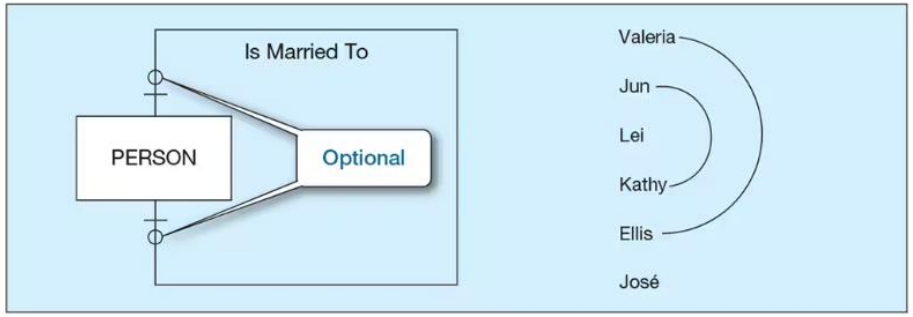
Examples of cardinality constraints



(a) Mandatory cardinalities



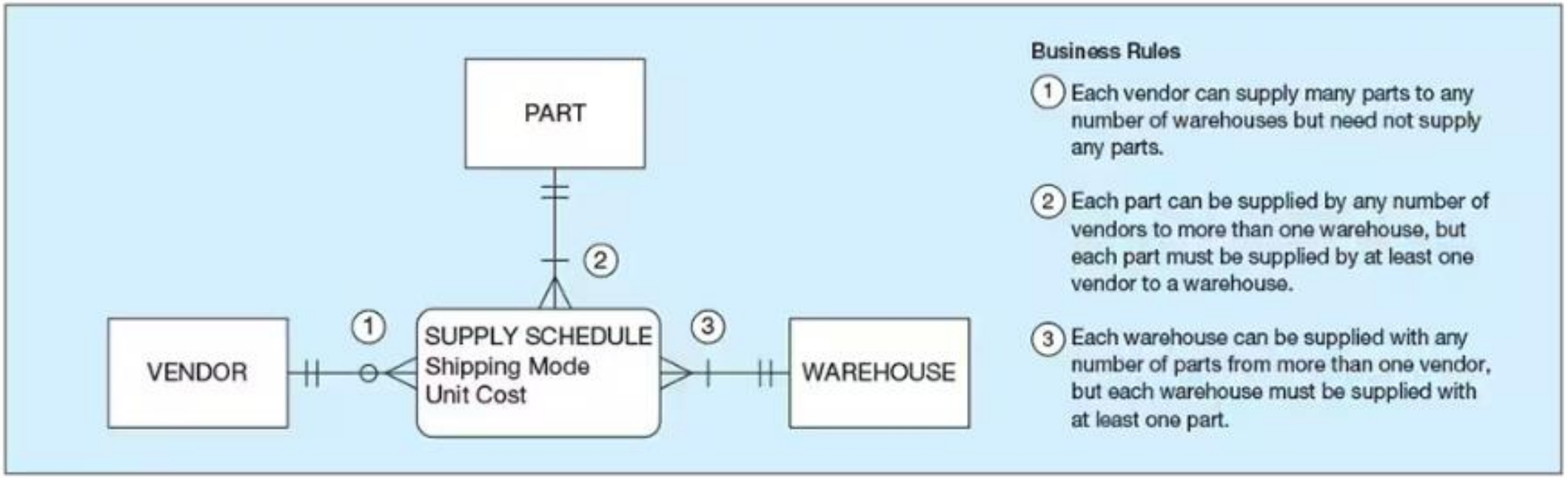
(b) One optional, one mandatory cardinality



(c) Optional cardinalities

Cardinalities on the ternary

Figure 2-18. Read the numbered rules on the right of the figure against the symbols.



- Next to each strong entity: always mandatory one. Every schedule names exactly one vendor, one part, one warehouse.
- A vendor may sit idle (optional many). Every part and every warehouse must appear at least once.

PART FOUR

Time, Pine Valley, and the road ahead

When history matters

Figure 2-19. A time stamp is a time value attached to a data value.

- Only today's price? A single-valued attribute is enough
- Accounting wants every price and when it took effect: {Price History (Effective Date, Price)}
- Regulation pushes history. Sarbanes-Oxley, HIPAA. So does analytics.
- History pushes toward many-to-many plus associative entities, at the cost of looser rules the database can no longer enforce
- If history might ever be required, consider it now

Figure 2-19

Simple example of time stamping

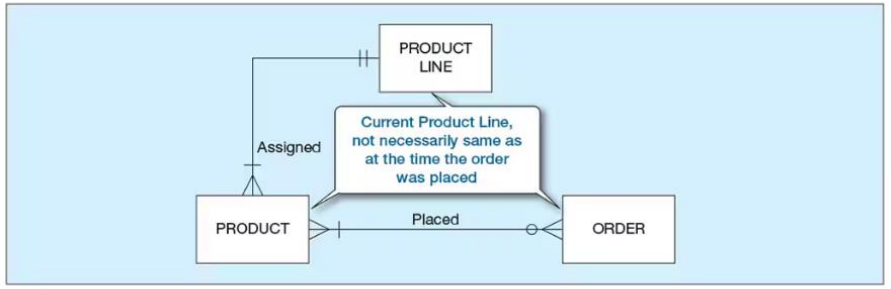


Time at Pine Valley: three tries

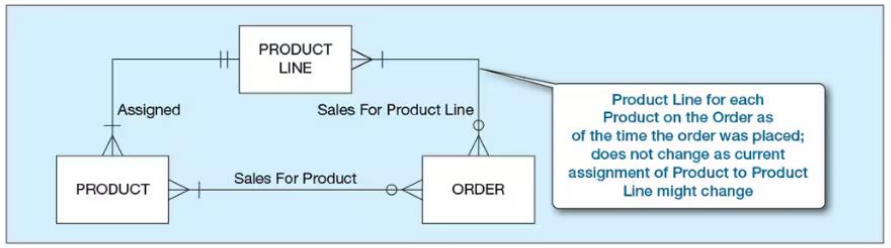
Figure 2-20. (a) loses history when a product changes lines. (b) credits the line in effect at sale time. (c) records every assignment with dates, but allows overlap.

Figure 2-20

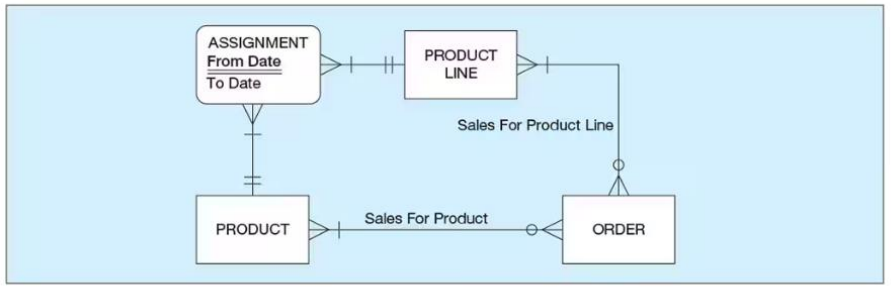
Example of time in Pine Valley Furniture product database



(a) E-R diagram not recognizing product reassignment



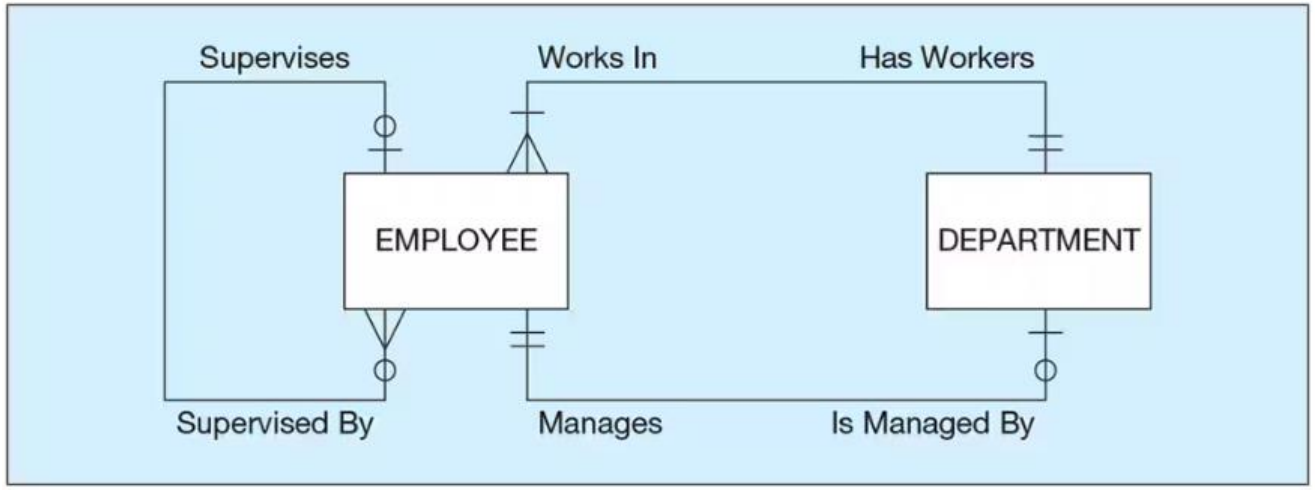
(b) E-R diagram recognizing product reassignment



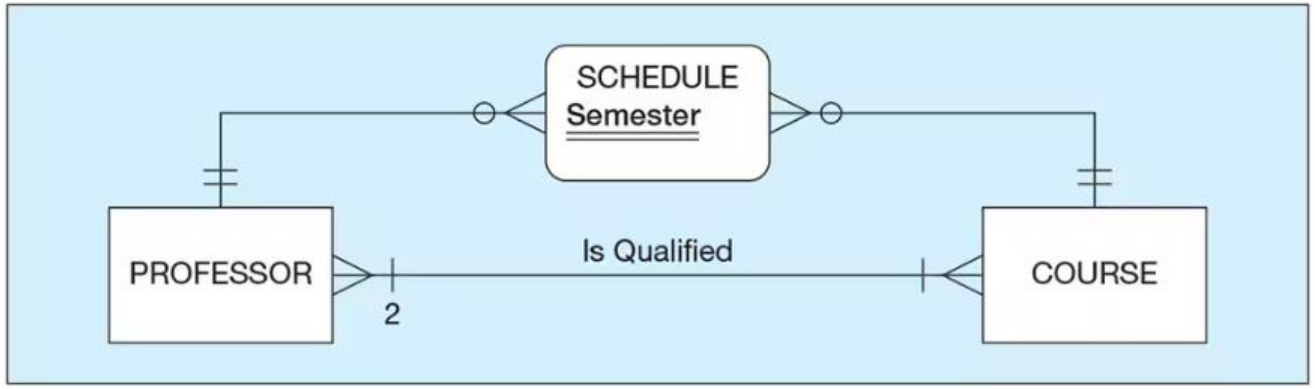
(c) E-R diagram with associative entity for product assignment to product line over time

Two entities, several relationships

Figure 2-21. Works In, Manages, and the unary Supervises all at once. The 2 is a fixed minimum: no course belongs to one professor. SCHEDULE's full identifier is professor plus course plus Semester.



(a) Employees and departments



(b) Professors and courses (fixed lower limit constraint)

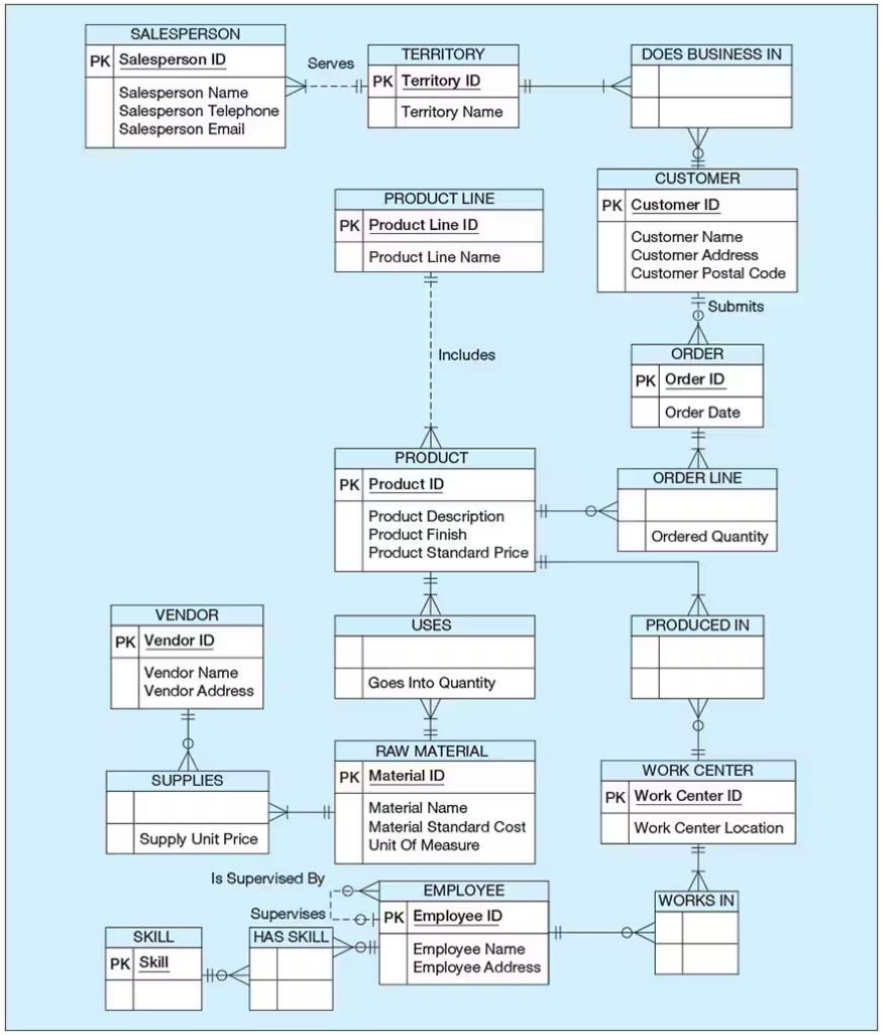
Naming relationships

- A present-tense verb phrase. Assigned To, Supplies, Teaches
- Name the action, not its result: Assigned To, never Assignment
- Nothing vague. Has and Is Related To say almost nothing.
- The definition carries the rest. Optional participation, fixed maxima and their reasons, exclusivity, how much history, whether participation can transfer.

Entities	Attributes	Relationships
Singular nouns in CAPITALS CUSTOMER, ORDER	Capitalized noun phrases Order Date, Customer ID	Present-tense verb phrases Submits, Assigned To

The Pine Valley model, complete

Figure 2-22, in Visio notation. Every construct from tonight, on one page.



- Find them. Seven associative entities: ORDER LINE, DOES BUSINESS IN, USES, SUPPLIES, PRODUCED IN, WORKS IN, HAS SKILL
- One unary relationship: Is Supervised By and Supervises
- Attributes riding an association: Ordered Quantity, Goes Into Quantity, Supply Unit Price
- Dashed line: the child's identifier does not include the parent's
- Read a few aloud. A product belongs to exactly one line. A customer may submit no orders at all. A work center must hold at least one employee.

Does the model answer real questions?

We check a design by walking real requests against it. SQL arrives properly in Chapter 5; you can already read this.

```
SELECT *
FROM Product
WHERE ProductDescription LIKE 'Computer Desk%';
```

PRODUCTID	PRODUCTDESCRIPTION	PRODUCTFINISH	PRODUCTSTANDARDPRICE
3	Computer Desk 48”	Oak	375.00
8	Computer Desk 64”	Pine	450.00

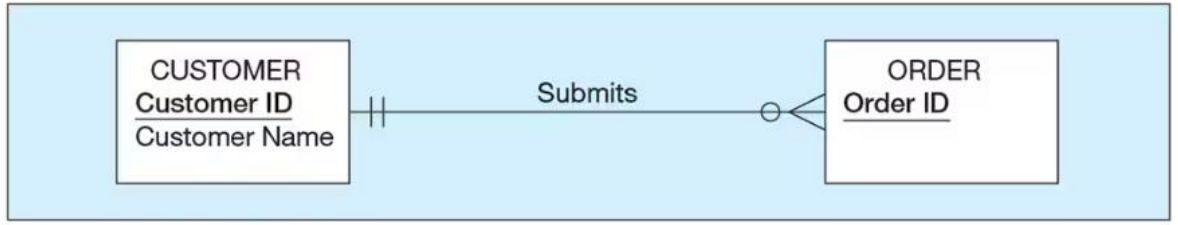
- One entity holds everything the salesperson asked for. Good sign.
- The percent sign matches anything that follows
- The book prints double quotes around text. DB Fiddle wants single quotes, as shown here.

User views: a named slice of the model

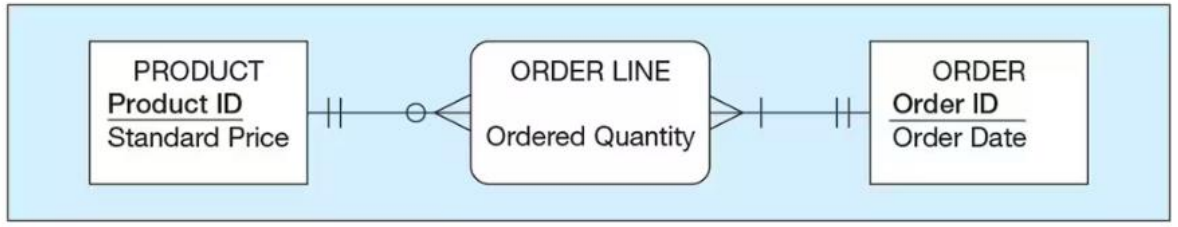
Figure 2-23. Each user sees only the entities and attributes a task needs.

Figure 2-23

Two user views for Pine Valley Furniture



(a) User View 1: Orders for customers



(b) User View 2: Orders for products

```
SELECT COUNT(OrderID)
FROM OrdersForCustomers
WHERE CustomerName =
      'Value Furniture';
```

- With the view, a two-entity question becomes a one-table query
- Answer: 4 orders
- Without it, a subquery finds the ID first. Chapter 6.

Before next class

- Tonight's worksheet. Finding Chapter 2 in Live Data, in DB Fiddle against the full Pine Valley database. Turn it in for participation.
- Study guide. The Chapter 2 guide on Blackboard walks every figure from tonight, with self-check questions.
- Blackboard tasks. Chapter 2 quiz, matching exercise, and the written assignment. Graduate students add the extension.
- Draw.io habit. Read your diagram aloud, both directions, before you submit anything.
- Exam 1 is September 23. Chapters 1 through 4.

Next: Chapter 3, the enhanced E-R model. Supertypes and subtypes settle the “who is a student” argument.